

EMCH 792: Optimal State Estimation
Homework 1
Written Portion Solutions

Instructor: Dr. Yimin Zhang, Department of Mechanical Engineering
University of South Carolina
Solutions by: JC Vaught

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Problem 1.2: Commuting Matrices

Problem Statement

Find two 2×2 matrices A and B such that $A \neq B$, neither A nor B are diagonal, $A \neq cB$ for any scalar c , and $AB = BA$. Find the eigenvectors of A and B . Note that they share an eigenvector. Interestingly, every pair of commuting matrices shares at least one eigenvector.

Step

Part (a): Finding Commuting Matrices

We seek two 2×2 matrices A and B such that $A \neq B$, neither is diagonal, $A \neq cB$ for any scalar c , and $AB = BA$.

Consider upper triangular matrices with the same diagonal entries:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad (1)$$

Step

Verification of Commutativity

Compute AB :

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad (2)$$

Compute BA :

$$BA = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad (3)$$

Therefore, $AB = BA$. Also: $A \neq B$, neither is diagonal, and $A \neq cB$ for any scalar c .

Step**Part (b): Finding Eigenvectors**

For matrix $\mathbf{A}$: Solve $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0}$:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{bmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{bmatrix} = (1 - \lambda)^2 = 0 \quad (4)$$

Thus, $\lambda = 1$ is a repeated eigenvalue. For $\lambda = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (5)$$

This gives $v_2 = 0$, so the eigenvector is $\mathbf{v}_\mathbf{A} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

For matrix $\mathbf{B}$: Similarly, $\lambda = 1$ is a repeated eigenvalue, and $\mathbf{v}_\mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Results**Final Answers:****Commuting Matrices:**

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Shared Eigenvector:

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This demonstrates the fundamental result that commuting matrices share at least one eigenvector.

Problem 1.3: Matrix Identities

Problem Statement

Prove the three identities of Equation (1.26):

1. $(AB)^T = B^T A^T$
2. $(AB)^{-1} = B^{-1} A^{-1}$
3. $\text{Tr}(AB) = \text{Tr}(BA)$

Step

Identity 1: $(AB)^T = B^T A^T$

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Let $C = AB$.
By definition of matrix multiplication and transpose:

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj} \quad (6)$$

$$(AB)^T_{ij} = C_{ji} = \sum_{k=1}^n A_{jk} B_{ki} \quad (7)$$

For $B^T A^T$:

$$(B^T A^T)_{ij} = \sum_{k=1}^n B_{ik}^T A_{kj}^T = \sum_{k=1}^n B_{ki} A_{jk} = \sum_{k=1}^n A_{jk} B_{ki} \quad (8)$$

Comparing: $(AB)^T_{ij} = (B^T A^T)_{ij}$ for all i, j . Therefore: $(AB)^T = B^T A^T$

Step

Identity 2: $(AB)^{-1} = B^{-1} A^{-1}$

Assume A and B are invertible square matrices. We verify that $B^{-1} A^{-1}$ is the inverse of AB :

$$(AB)(B^{-1} A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \quad (9)$$

$$(B^{-1} A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \quad (10)$$

Since both products equal the identity: $(AB)^{-1} = B^{-1} A^{-1}$

Step

Identity 3: $\text{Tr}(\mathbf{A}\mathbf{B}) = \text{Tr}(\mathbf{B}\mathbf{A})$

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times m$.

$$\text{Tr}(\mathbf{A}\mathbf{B}) = \sum_{i=1}^m (\mathbf{A}\mathbf{B})_{ii} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ji} \quad (11)$$

$$\text{Tr}(\mathbf{B}\mathbf{A}) = \sum_{j=1}^n (\mathbf{B}\mathbf{A})_{jj} = \sum_{j=1}^n \sum_{i=1}^m B_{ji} A_{ij} \quad (12)$$

Since scalar multiplication is commutative and summation order can be exchanged:

$$\text{Tr}(\mathbf{B}\mathbf{A}) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ji} = \text{Tr}(\mathbf{A}\mathbf{B}) \quad (13)$$

Therefore: $\boxed{\text{Tr}(\mathbf{A}\mathbf{B}) = \text{Tr}(\mathbf{B}\mathbf{A})}$

Results

Summary of Proven Identities:

1. $\boxed{(AB)^T = B^T A^T}$ — Transpose of a product reverses order
2. $\boxed{(AB)^{-1} = B^{-1} A^{-1}}$ — Inverse of a product reverses order
3. $\boxed{\text{Tr}(AB) = \text{Tr}(BA)}$ — Trace is invariant under cyclic permutations

Problem 1.4: Trace Derivative

Problem Statement

Find the partial derivative of the trace of AB with respect to A .

Step

Setup

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times m$. The trace of $\mathbf{AB}$ is:

$$\text{Tr}(\mathbf{AB}) = \sum_{i=1}^m (\mathbf{AB})_{ii} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ji} \quad (14)$$

Step

Computing the Derivative

To find the derivative with respect to $\mathbf{A}$, compute the derivative with respect to each element A_{kl} :

$$\frac{\partial \text{Tr}(\mathbf{AB})}{\partial A_{kl}} = \frac{\partial}{\partial A_{kl}} \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ji} \quad (15)$$

Since only the term with $i = k$ and $j = l$ contains A_{kl} :

$$\frac{\partial \text{Tr}(\mathbf{AB})}{\partial A_{kl}} = B_{lk} \quad (16)$$

This is precisely the (k, l) element of $\mathbf{B}^T$.

Results

Final Answer:

$$\boxed{\frac{\partial \text{Tr}(\mathbf{AB})}{\partial \mathbf{A}} = \mathbf{B}^T}$$

The gradient of $\text{Tr}(\mathbf{AB})$ with respect to $\mathbf{A}$ is the transpose of $\mathbf{B}$.

Problem 1.6: Block Matrix Equation

Problem Statement

Suppose that A is invertible and

$$\begin{bmatrix} A & A \\ B & A \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

Find B and C in terms of A .

Step

Expanding the Block Matrix Multiplication

First row:

$$A \cdot A + A \cdot C = 0 \Rightarrow A^2 + AC = 0$$

Second row:

$$B \cdot A + A \cdot C = I \Rightarrow BA + AC = I$$

Step

Solving for C

From the first row: $AC = -A^2$

Since A is invertible, multiply both sides by A^{-1} on the left:

$$C = -A$$

Step

Solving for B

Substituting $C = -A$ into the second row:

$$BA + A(-A) = I \Rightarrow BA - A^2 = I \Rightarrow BA = I + A^2$$

Multiplying both sides by A^{-1} on the right:

$$B = (I + A^2)A^{-1} = A^{-1} + A$$

Step

Verification

Check: $BA = (A^{-1} + A)A = A^{-1}A + A^2 = I + A^2 \checkmark$

Results**Final Answers:**

$$B = A^{-1} + A$$

$$C = -A$$

Problem 1.7: Symmetric Matrix Product

Problem Statement

Show that AB may not be symmetric even though both A and B are symmetric.

Step

Counterexample Construction

We need to find symmetric matrices A and B such that $AB \neq (AB)^T$.

Let:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Step

Verifying Symmetry of A and B

$$A^T = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = A \quad \checkmark$$

$$B^T = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = B \quad \checkmark$$

Step

Computing the Product

$$AB = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Step

Comparison

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = (AB)^T$$

Therefore, $AB \neq (AB)^T$, demonstrating that AB is **not** symmetric.

Results

Conclusion:

Although both $\mathbf{A}$ and $\mathbf{B}$ are symmetric matrices, their product $\mathbf{AB}$ is **not symmetric**.

Counterexample:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{AB} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \neq (\mathbf{AB})^T$$

Note: The product $\mathbf{AB}$ is symmetric if and only if $\mathbf{AB} = \mathbf{BA}$, i.e., if $\mathbf{A}$ and $\mathbf{B}$ commute.